

- 2 (a) (i) $pV = nRT$ C1
 $V = (8.31 \times 300)/(1.02 \times 10^5)$ C1
 $= 0.0244 \text{ m}^3$ (if uses Celsius, then 0/2) A1 [2]
- (ii) volume occupied by one atom $= 0.0244 / (6.02 \times 10^{23}) = 4.06 \times 10^{-26} \text{ m}^3$ M1
separation $\approx \sqrt[3]{(4.06 \times 10^{-26})}$ A1
 $= 3.44 \times 10^{-9} \text{ m}$ A0 [2]
- (b) (i) $F = GMm / r^2$ C1
 $= (6.67 \times 10^{-11} \times \{4 \times 1.66 \times 10^{-27}\}^2) / (3.44 \times 10^{-9})^2$ C1
 $= 2.49 \times 10^{-46} \text{ N}$ A1 [3]
- (ii) ratio $= (4 \times 1.66 \times 10^{-27} \times 9.8) / 2.49 \times 10^{-46}$ C1
 $= 2.6 \times 10^{20}$ A1 [2]
- (c) assumption that forces between atoms are negligible B1
comment e.g. ratio shows gravitational force to be very small
e.g. force is very much less than weight
e.g. if there are forces, they are not gravitational B1 [2]

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2007	9702	04